

Mr. James Duncan

Lead Backend Engineering | 9.4+ Years Experience

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Executive Summary

Engineering leader with deep technical background in Kotlin, Node.js, and Go. 9 years of experience delivering enterprise-scale solutions.

Leadership Experience

AI Engineer 2024-05 – Present

OpenAI

- Reduced infrastructure costs by 36% through optimization using Replicate
- Developed real-time data processing pipeline using Ansible, handling 5000+ transactions/second
- Reduced infrastructure costs by 62% through optimization using Ansible
- Led migration from monolithic to microservices using Ansible, improving system scalability by 23%

Platform Engineer 2020-06 – 2024-05

DataPulse

- Architected and built RESTful APIs serving 5000+ requests/day using Express.js, reducing response time by 68%
- Architected and built RESTful APIs serving 100+ requests/day using Ansible, reducing response time by 69%

AI Engineer 2017-03 – 2020-05

Supabase

- Created authentication and authorization system using GitHub Actions, securing 100+ user accounts
- Mentored team of 8 junior developers on RabbitMQ best practices
- Designed and implemented message queue system using RabbitMQ, processing 500+ messages/hour

AI Engineer 2016-09 – 2017-03

OpenAI

- Optimized database queries using Rails, improving query performance by 20%
- Developed real-time data processing pipeline using Cassandra, handling 5000+ users/second
- Built event-driven architecture using GraphQL, processing 10000+ events daily
- Led migration from monolithic to microservices using Rails, improving system scalability by 40%

Education

B.A. in Computer Science — University of Waterloo, 2017

Ph.D. in Electrical Engineering — University of Waterloo, 2014

Key Skills

Kotlin | Node.js | Go | PHP | Express.js | ASP.NET | Actix | LlamaIndex | SQLite | Supabase | PostgreSQL |
DynamoDB | CockroachDB | GitHub Actions | ArgoCD | RabbitMQ | Git | gRPC | Fly.io | GCP | Replicate |
Azure OpenAI | Google Gemini | Microservices | Event-Driven Architecture | LLM Integration

Credentials

Google Cloud Professional Cloud Developer — Google, 2025